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# Markdown Renderer Test 2: Intermediate

Tests more languages, richer math environments, more Mermaid diagram types, and features that often live alongside them (tables, footnotes, task lists).

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## 1. Code Highlighting (Many Languages)

**Rust**

```rust
use std::collections::HashMap;

#[derive(Debug, Clone)]
struct User {
    id: u32,
    name: String,
}

fn main() {
    let mut users: HashMap<u32, User> = HashMap::new();
    users.insert(1, User { id: 1, name: "Ada".into() });
```

```rust
    if let Some(u) = users.get(&1) {
        println!("Found {:?}", u);
    }
}
```

**TypeScript**

```typescript
interface Point<T extends number = number> {
  x: T;
  y: T;
}

export function distance(a: Point, b: Point): number {
  return Math.hypot(a.x - b.x, a.y - b.y);
}
```

**SQL**

```sql
SELECT u.name, COUNT(o.id) AS order_count
FROM users u
LEFT JOIN orders o ON o.user_id = u.id
WHERE u.created_at >= '2025-01-01'
GROUP BY u.name
HAVING COUNT(o.id) > 5
ORDER BY order_count DESC;
```

**Go**

```go
package main

import "fmt"

func main() {
    ch := make(chan int)
    go func() { ch <- 42 }()
    fmt.Println(<-ch)
}
```

**HTML + CSS**

```html
<div class="card" id="main">
  <h1 style="color: #333;">Title</h1>
  <!-- a comment -->
  <a href="https://example.com">Link</a>
</div>
```

```css
.card > h1:hover {
  color: rgb(255 0 128 / 80%);
  transition: all 0.3s ease-in-out;
}
```

### YAML and TOML

```yaml
services:
  web:
    image: nginx:latest
    ports:
      - "8080:80"
    environment:
      - DEBUG=true
```

```toml
[package]
name = "demo"
version = "0.1.0"

[dependencies]
serde = { version = "1.0", features = ["derive"] }
```

### Diff

```diff
- const old = "removed line";
+ const updated = "added line";
  const unchanged = true;
```

### Line with no language

```
Plain text block.
    Indentation   and   spacing   must   be   preserved.
No highlighting expected here.
```

---

## 2. Links

| Type | Example |
| --- | --- |
| Inline | Wikipedia |
| URL with parentheses | Function (mathematics) |
| URL with query | Search |
| Link with code | `npm install` |
| Bold link | **Important** |
| Image link |  |

Footnote reference here[1] and another one[2].

---

[1] This is the first footnote, with a link.

[2] A named footnote with `inline code`.

## 3. LaTeX Math

**Inline variety**
Inline examples: $\alpha + \beta = \gamma$, $x_i^2$, $\sqrt[3]{27} = 3$, $\vec{v} \cdot \vec{w}$, $\mathbb{R}^n$, and $\lim_{x \to 0} \frac{\sin x}{x} = 1$.

**Aligned equations**

$$\begin{aligned} (a+b)^2 &= a^2 + 2ab + b^2 \\ (a-b)^2 &= a^2 - 2ab + b^2 \\ (a+b)(a-b) &= a^2 - b^2 \end{aligned} \tag{1}$$

**Matrices**

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}, \quad \det(A) = \begin{Vmatrix} a & b \\ c & d \end{Vmatrix} = ad - bc \tag{2}$$

**Piecewise function**

$$f(x) = \begin{cases} x^2 & \text{if } \ x \geq 0 \\ -x & \text{if } \ x < 0 \end{cases} \tag{3}$$

**Maxwell's equations**

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \frac{\rho}{\varepsilon_0} \\ \nabla \cdot \mathbf{B} &= 0 \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \times \mathbf{B} &= \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \end{aligned} \tag{4}$$

**Bayes and probability**

$$P(A \mid B) = \frac{P(B \mid A)\, P(A)}{P(B)} \tag{5}$$

---

## 4. Mermaid (Many Diagram Types)

### Class diagram

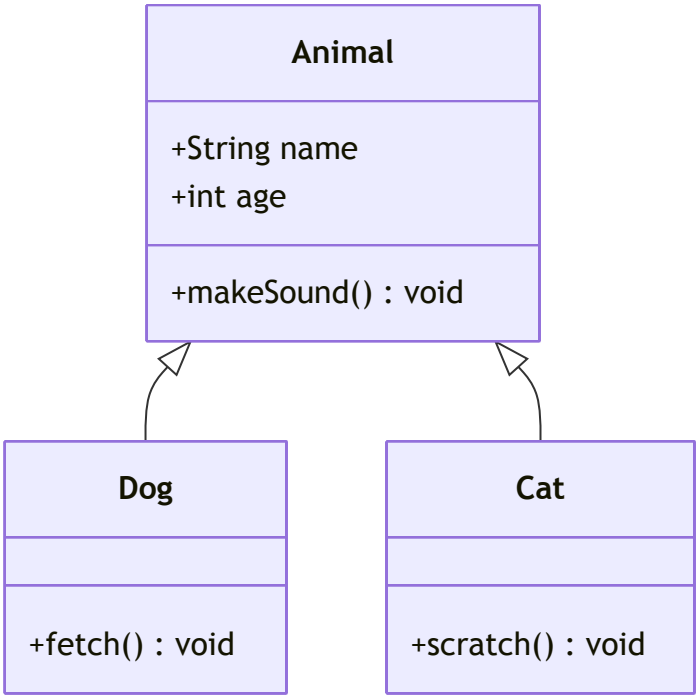



*Figure 1: Mermaid diagram 1*

### State diagram

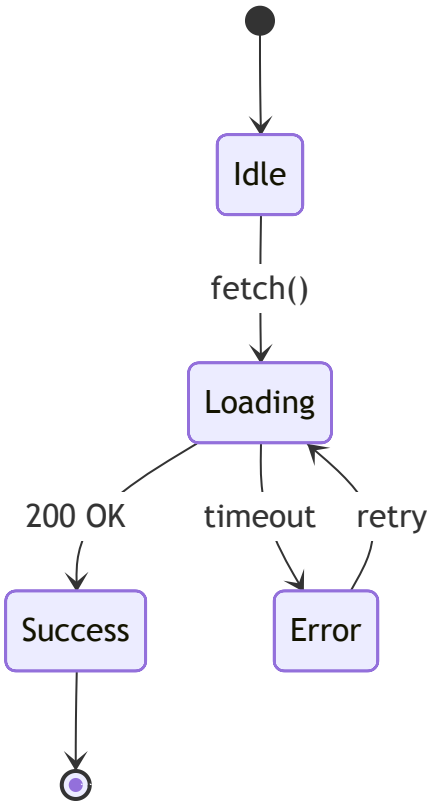



*Figure 2: Mermaid diagram 2*

### Gantt chart

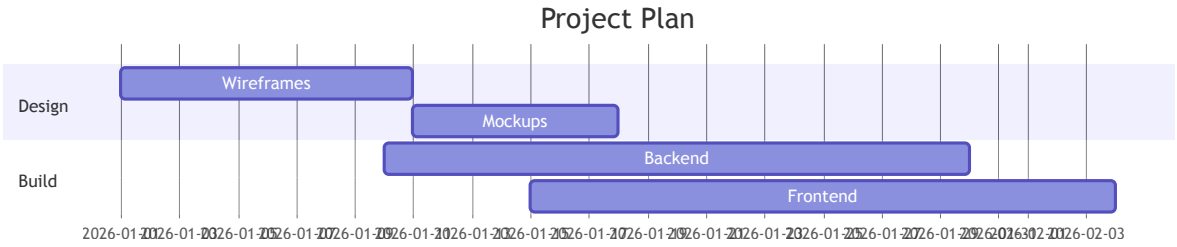



*Figure 3: Mermaid diagram 3*

### Pie chart

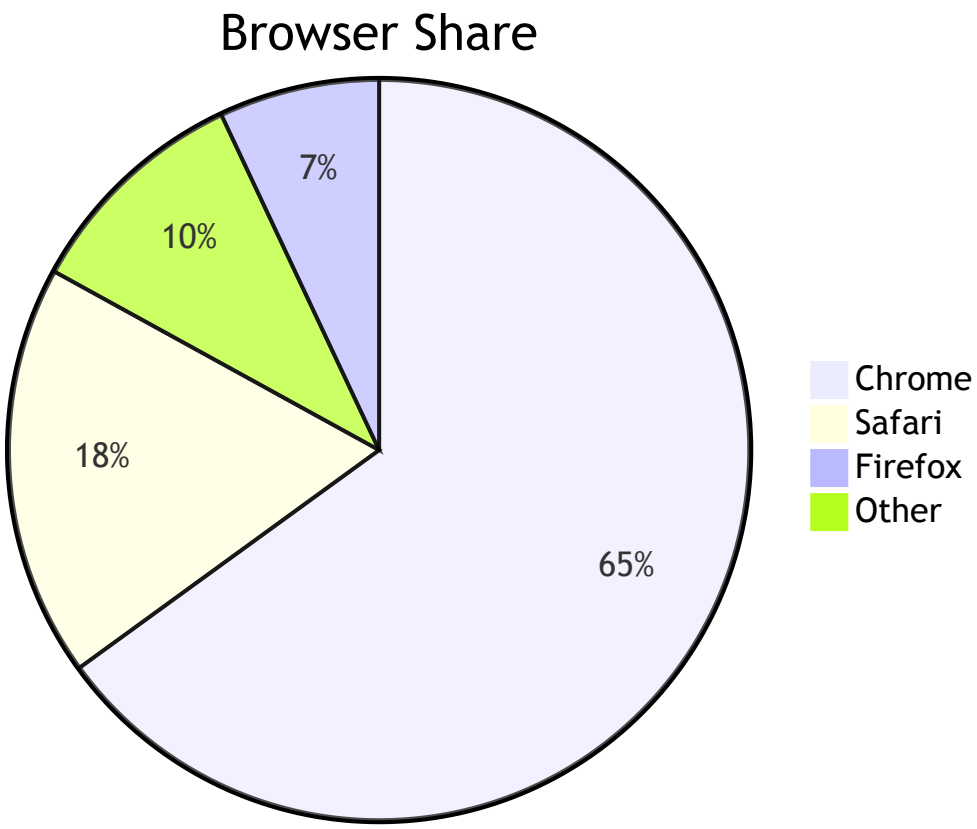



*Figure 4: Mermaid diagram 4*

**Entity relationship**

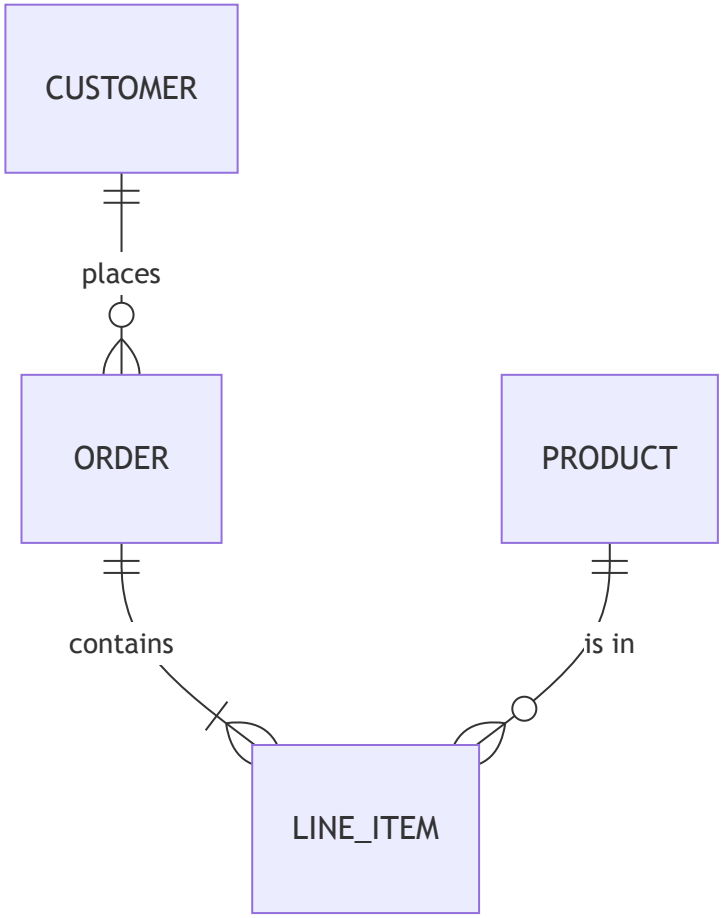



*Figure 5: Mermaid diagram 5*

---

## 5. Misc Features

- ☒ Completed task
- ☐ Pending task
- ~~Strikethrough text~~
- $H_2O$ and $x^2$ (extended syntax; may not render everywhere)

> **Note:** a blockquote with `code`, a link, and math $E = mc^2$.

---

## Expected Results

- ☐ Every language block has distinct token coloring; the diff shows red/green lines
- ☐ The no-language block stays monospaced with spacing intact
- ☐ The Wikipedia link with parentheses is not cut off at the first `)`
- ☐ Footnotes render with back-references
- ☐ Aligned equations line up on the `=` sign; matrices have brackets
- ☐ The piecewise function has a large left brace

- ☐ Five Mermaid diagrams render: class, state, Gantt, pie, ER
- ☐ Math inside the blockquote renders